

Reinforcement with Deep Learning

a Discussion

Guillaume.Lozenquez

@imt-nord-europe.fr



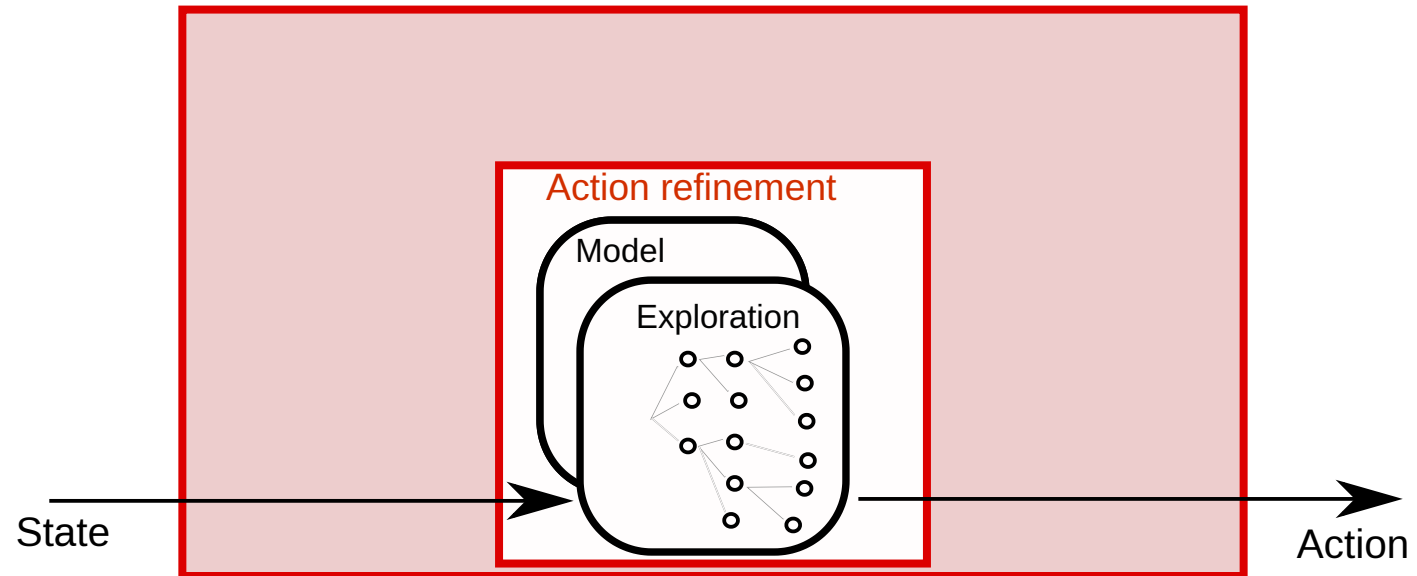
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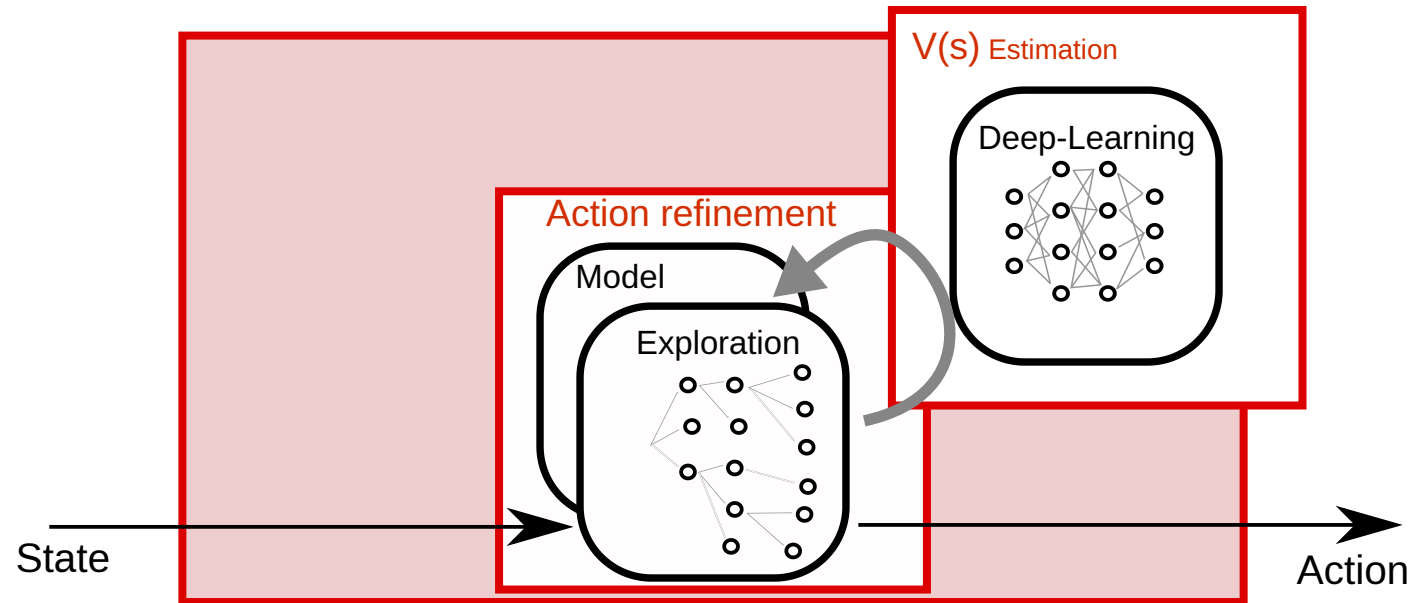
Basic approach - Random Exploration Tree

Explore locally possible evolution from the current state.



Basic approach - Incorporate Neural Networks

Incorporate Intuition as Deep Learning based Estimation



- Cut exploration branches based on a heuristic value function

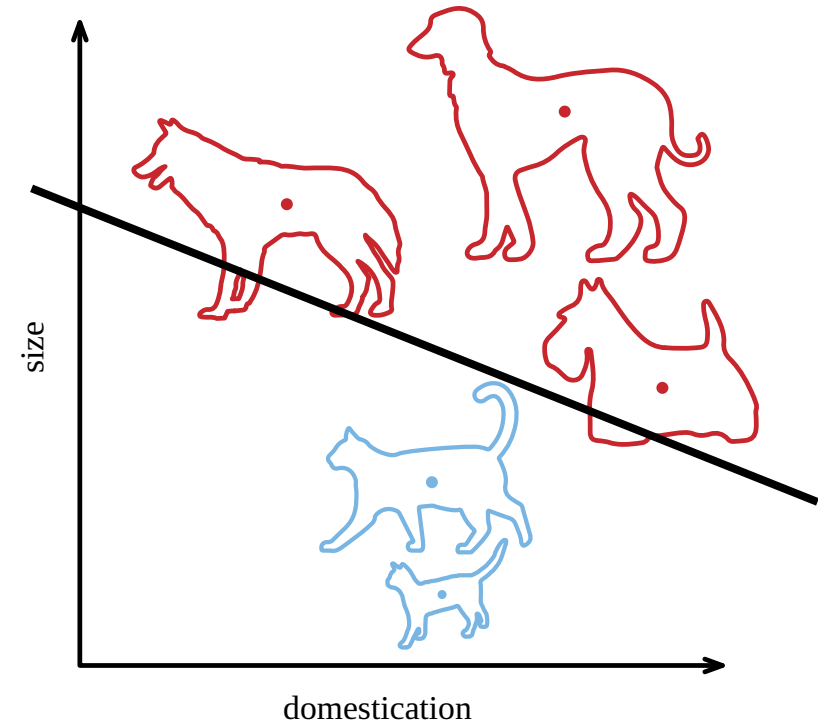
Le Deep-Learning: un cerveau à un neurone

Le perceptron:

$$si : (p_1 \cdot x + p_2 \cdot y > b)$$

Les paramètres:

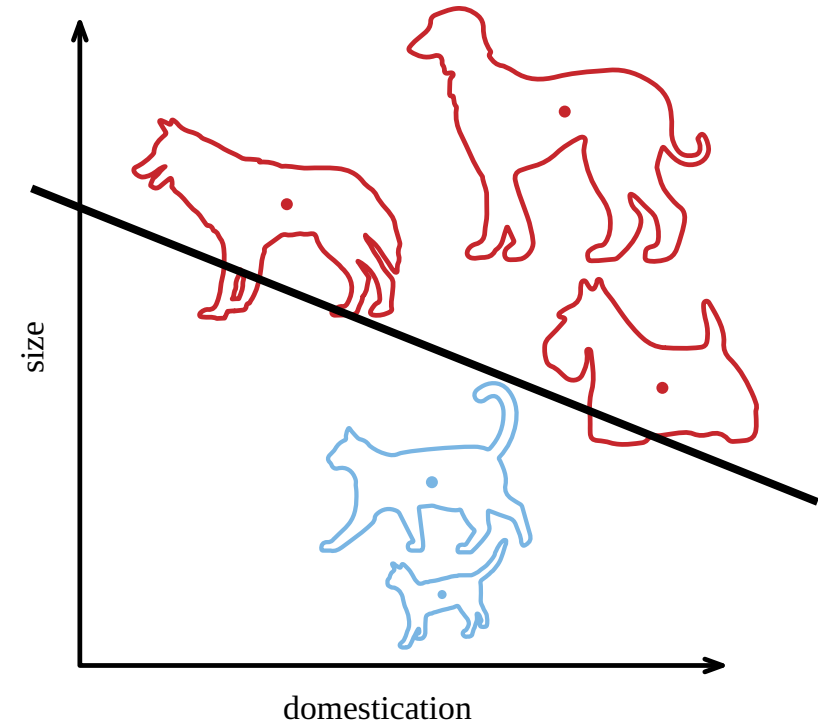
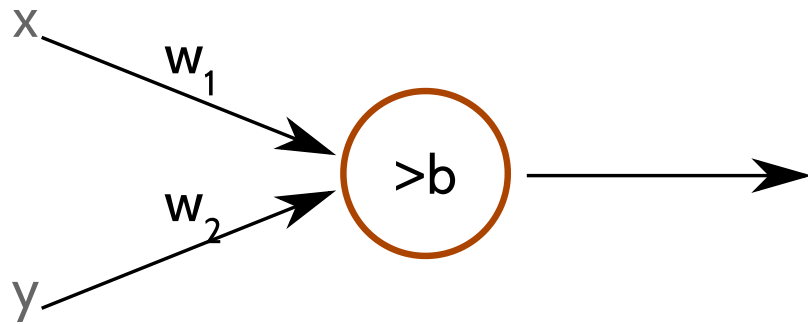
- ▶ les poids: p_1, p_2
- ▶ un seuil: b



Le grand problème des data-sciences: Classifier

Le Deep-Learning: un cerveau à un neurone

Le perceptron:



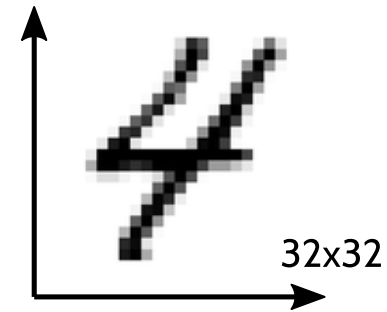
Le grand problème des data-sciences: Classifier

Le Deep-Learning: des réseaux de neurones profonds

► La rétropropagation:

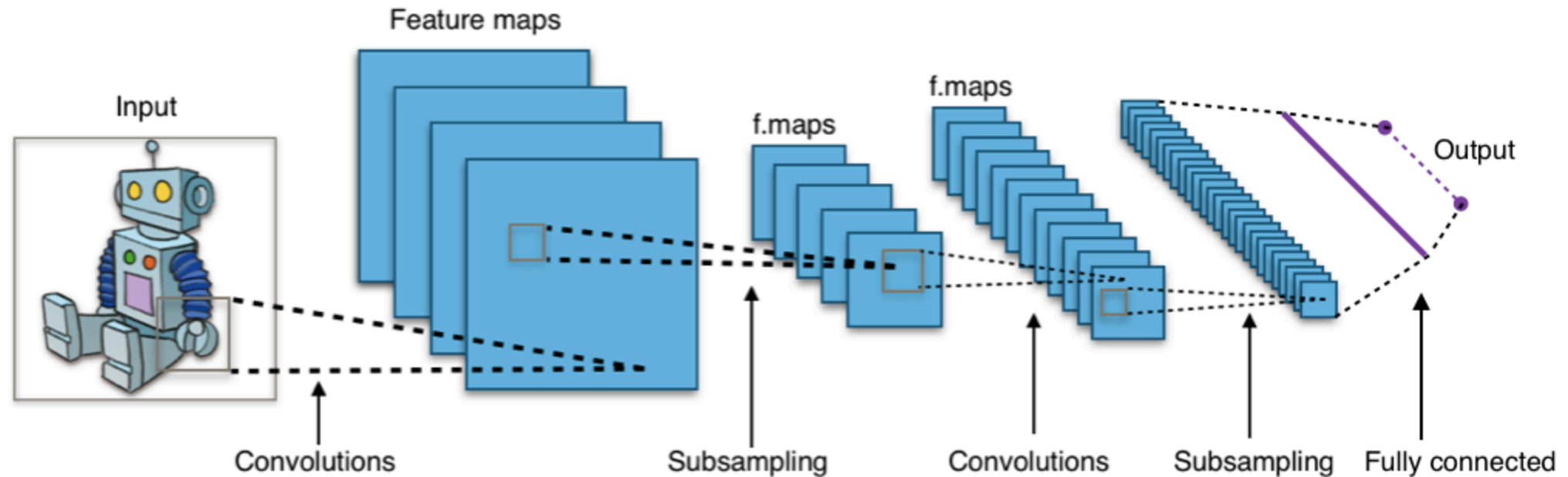
LeNet (1989-1998) *Yan LeCun et al.*

Livre: *Quand la machine Apprend*



Le Deep-Learning: des réseaux de neurones profonds

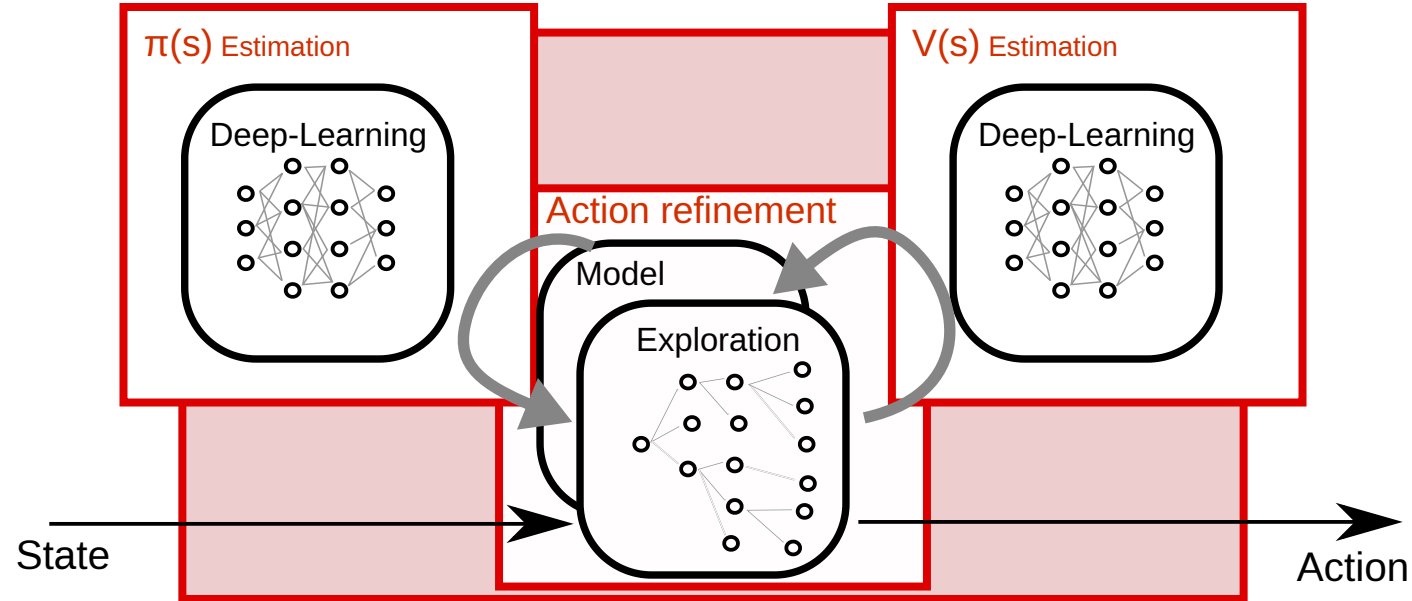
Une démocratisation de l'usage des CNN



La classification: pour n'importe qui, avec des donnés.

La "killing Architecture" - AlphaZero

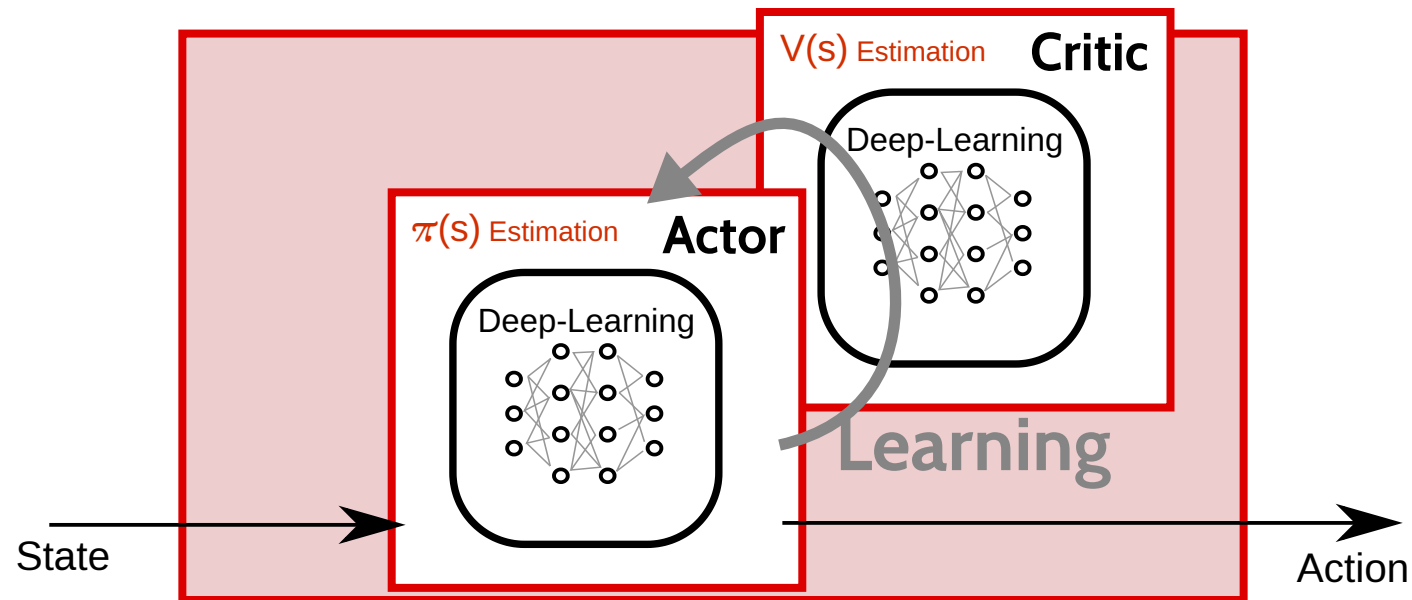
Use both policy and value estimation.



1. Iterative process: (Play phase // Analysis phase)
2. Suppose a model of the systems (game)

Le Deep Reinforcement Learning

Both policy and value estimation, but in Reinforcement Learning approach



Classical *Actor-Critic architecture* (see PPO algorithm for example).